



Mirpur University of Science & Technology
Department of Software Engineering

LAB TASK 3
SOFTWARE QUALITY ENGINEERING

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Submitted to: Engr. Asim Javed

Section: A

1. Test Plan Identifier

TEST PLAN	University Library Management System
Document ID	LMS-TP-001
Version	1.0
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Reviewed By	Engr. Asim Javed
Status	Report

2. Introduction

What is the LMS? What is its purpose and who are its users?

The University Library Management System (LMS) is a web-based software application developed to automate and manage library operations within a university environment. The system allows students to search books, borrow library resources, manage their accounts, and receive notifications, while library staff can manage inventory, loans, and user records efficiently.

What is the scope of this test plan? Which modules are covered? What version is under test?

This test plan defines the strategy, scope, resources, schedule, and testing activities for LMS Version 1.0. The objective of testing is to verify that the system fulfills specified functional and performance requirements and provides reliable services to users before deployment.

Key assumptions and constraints that affect testing

This test cycle specifically focuses on Module 1 (Authentication) and Module 2 (Book Search). The plan covers login validation, session management, search functionality, filtering, pagination, and response-time verification. Loan Management and Notification modules are intentionally excluded from this phase and will be tested in future lab sessions.

The testing approach follows IEEE 829 standards to ensure structured documentation, traceability, consistency, and professional software quality assurance practices.

3. Test Items

Item	Version / Build	Description
LMS Frontend	v1.0.0 Build #42	React.js-based user interface for students and library staff
LMS Backend	v1.0.0	Node.js and Express REST API server
LMS Database	PostgreSQL v14.2	Stores student, loan, book, and authentication data
Authentication Module	v1.0.0	Handles login, sessions, password reset, and security
Book Search Module	v1.0.0	Supports searching, filtering, and pagination
Test Dataset	LMS-TESTDATA-001	Preloaded database with books and test accounts
JIRA Workspace	SCRUM Project Board	Used for test management and defect tracking

4. Features to be tested

Feature	Requirement ID	Priority
User login using valid credentials	REQ-01	High
Invalid login handling	REQ-01	High
Account lockout after 3 failed attempts	REQ-02	High
Account unlock verification	REQ-02	Medium
Password reset email generation	REQ-03	Medium
Password reset email delivery within 2 minutes	REQ-03	Medium
Session timeout after inactivity	REQ-04	High

Session timer reset on user activity	REQ-04	Medium
Search books by title	REQ-05	High
Search books by author	REQ-05	High
Search books by ISBN	REQ-05	High
Filter books by genre	REQ-06	Medium
Filter books by availability	REQ-06	Medium
Search result response time under 2 seconds	REQ-07	High
Pagination for result sets exceeding 20 items	REQ-08	Low

5. Features NOT to be Tested

Feature Excluded	Justification
Loan Management Module (REQ-09 to REQ-12)	Planned for future testing cycle in Week 4 lab sessions
Notification Module (REQ-13 to REQ-16)	Requires SMS gateway and advanced email configuration not available in current environment
Mobile Responsiveness Testing	LMS Version 1.0 is desktop-focused only
Cross-Browser Compatibility Testing	Current testing limited to Google Chrome to maintain environment consistency
API Penetration Testing	Specialized security tools and penetration testing techniques will be covered in later labs
Load and Stress Testing	JMeter/load-testing infrastructure not configured during this lab
Database Backup and Recovery Testing	Infrastructure-level testing handled by DevOps team

6. Test Approach

The LMS system will undergo system-level black-box testing to verify compliance with defined requirements. Testing will primarily focus on functional behavior, security-related validation, and basic performance verification.

Manual testing techniques will be used during this phase because the objective of the lab is to establish testing documentation and structured test management processes before automation activities begin.

The following testing approaches will be applied:

Aspect	Testing Approach
Testing Level	System Testing
Testing Type	Functional, Security, Performance
Execution Method	Manual Testing
Black-Box Techniques	Equivalence Partitioning and Boundary Value Analysis
Defect Tracking	JIRA Software
Regression Testing	Smoke tests executed after defect fixes
Test Prioritization	High-priority requirements tested first
Test Data Management	Preloaded LMS test dataset restored after each session

Boundary testing will be especially important for session timeout duration, failed login attempts, and response-time validation. Equivalence partitioning will be used for valid and invalid input scenarios such as usernames, passwords, and search queries.

Defects identified during execution will be logged in JIRA with severity classification and supporting evidence.

7. Pass / Fail Criteria

Level	Pass Criteria	Fail Criteria
Test Case Level	Actual result exactly matches expected result	Any mismatch between actual and expected result
Functional Testing	All mandatory functions execute successfully	Core functionality unavailable or incorrect
Performance Testing	Search results displayed within 2 seconds	Response exceeds defined limit
Security Validation	Account lockout and session timeout work correctly	Security controls fail or bypass possible
Test Cycle Exit Criteria	100% High-priority tests executed; no Critical defects open	Critical defects unresolved or incomplete execution

Defect Severity Classification

Severity	Description
Critical	System crash, data corruption, or complete module failure
High	Major functionality broken without workaround
Medium	Partial feature malfunction with workaround available
Low	Cosmetic or minor UI issue

8. Suspension Criteria & Resumption Requirements

Suspension Trigger	Threshold	Resumption Requirement
LMS staging server unavailable	More than 2 hours downtime	Server restored and verified stable
Critical authentication defect	1 Critical defect open	Fix deployed and re-tested successfully
Test database corruption	Invalid or inconsistent test data	Database restored from backup snapshot
Smoke test failure rate exceeds 30%	More than 30% failure in core tests	Stable build provided by development team
Excessive High-severity defects	More than 20% High defects in sprint	Defect triage completed and fixes deployed

Testing activities will resume only after environment stability and functionality are verified.

9. Test Deliverables

Deliverable	Description	Format	Due Date
Requirement Traceability Matrix	Mapping of requirements to test cases	XLSX	Week 3
IEEE 829 Test Plan	Formal testing strategy document	PDF	Week 3
Test Case Specifications	Detailed manual test cases	JIRA	Week 4
Test Data Specification	Test datasets and setup information	XLSX	Week 4

Test Execution Report	Test execution results and summaries	PDF	Week 5
Defect Report	Defect logs maintained in JIRA	CSV/JIRA	Week 5
Test Closure Report	Final testing summary and lessons learned	PDF	Week 6

10. Environmental Needs

Category	Requirement
Operating System	Windows 10/11 or Ubuntu 22.04
Processor	Intel i5 or equivalent
RAM	Minimum 8 GB
Browser	Google Chrome v120+
Internet	Stable broadband connection
Test Environment	LMS staging server
Database	PostgreSQL v14.2
Test Management Tool	JIRA Software
Email Testing Tool	Gmail test account
Timing Tool	Stopwatch or browser timer

11. Responsibilities

Role	Assigned To	Responsibilities
Test Manager	Maryam Ikhlaq	Manage testing process, review progress, maintain documentation
Test Designer	Maryam Ikhlaq	Design test cases and prepare test data
Test Executor	Maryam Ikhlaq	Execute test cases and report defects

Development Team	LMS Developers	Fix defects and provide stable builds
Environment Administrator	Instructor / DevOps	Maintain staging environment and restore data
Reviewer	Engr. Asim Javaid	Review and approve test documentation
Stakeholder	University Library Staff	Clarify business requirements and validate workflows

12. Schedule

Week	Phase	Activity	Deliverable
Week 3	Requirements & Planning	RTM creation and IEEE 829 Test Plan preparation	RTM + Test Plan
Week 4	Test Design	Create detailed test cases for M1 and M2	Test Case Specification
Week 5	Test Execution	Execute Authentication Module tests	Execution Report
Week 6	Test Execution	Execute Book Search Module tests	Execution Report
Week 7	Defect Management	Retesting and regression testing	Defect Closure Report
Week 8	Test Closure	Final reporting and documentation archive	Test Closure Report

URL for Jira:

<https://maryam-ikhlaq.atlassian.net/jira/software/projects/SCRUM/boards/1>